STEM LABS CURRICULUM

Section 1: Junior Builders (Grades 1 & 2)

1. Make Your Own Spectacles

Learning Objective: Understand symmetry and facial ergonomics.

Activity: Using cardboard or 3D pen templates to design wearable glasses.

Real-World Link: Optical engineering and fashion design.

2. Hanger Hub

Learning Objective: Understand balance and structure.

Activity: Create a hanger system using simple materials.

Real-World Link: Home organization systems.

3. Make Your Own Gear

Learning Objective: Understand basic gear concepts.

Activity: Build simple gears using cardboard.

Real-World Link: Mechanical engineering basics.

4. Gear Mechanisms

Learning Objective: Learn how gears transmit motion.

Activity: Assemble multiple gears and observe movement.

Real-World Link: Machines and automotive systems.

5. Windmill

Learning Objective: Understand energy conversion.

Activity: Build a working windmill model.

Real-World Link: Renewable energy systems.

6. Giant Wheel Mini

Learning Objective: Understand rotational motion.

Activity: Create a mini giant wheel model.

Real-World Link: Amusement park engineering.

7. Dancing Robot

Learning Objective: Understand basic motion and vibration.

Activity: Build a simple dancing robot.

Real-World Link: Entertainment robotics.

8. Rolling Car

Learning Objective: Understand movement and wheels.

Activity: Design a rolling car.

Real-World Link: Automobile basics.

9. Interfacing Sensors

Learning Objective: Understand sensor usage.

Activity: Connect simple sensors and observe outputs.

Real-World Link: Smart devices.

10. Interfacing Arduino

Learning Objective: Understand basic coding and electronics.

Activity: Connect Arduino and run simple programs.

Real-World Link: Automation and IoT.